

Stats 21 Drill — Answer Key

15 questions targeting the 10 you missed on the practice final. Every answer verified by running the code.
Themes: broadcasting (valid & invalid), "does it really error?", list mutation, return, any/all truthiness, bool dtype, argmax.

Quick grid

Q1: a	Q2: c	Q3: a	Q4: b	Q5: b
Q6: c	Q7: a	Q8: b	Q9: a	Q10: b
Q11: b	Q12: a	Q13: a	Q14: c	Q15: b

Detailed answers

Q1. Answer: a

```
Output:  
[[11 12 13]  
 [24 25 26]]
```

Why: (2,1) column vector broadcasts across the columns: row 0 gets +10, row 1 gets +20.

Q2. Answer: c

```
Output:  
ValueError
```

Why: (3,2) + (3,): lined up from the right it's 2 vs 3 — mismatch, so ValueError.

Q3. Answer: a

```
Output:  
[[11 22]  
 [13 24]  
 [15 26]]
```

Why: (2,) matches the 2 columns and is added to every row.

Q4. Answer: b

```
Output:  
['a']  
['a', 'b']
```

Why: The default list is created once and persists, so it accumulates: ['a'] then ['a','b']. No error — lists are legal defaults.

Q5. Answer: b

```
Output:  
ok
```

Why: return "ok" exits the function immediately; the print after it never runs.

Q6. Answer: c

```
Output:  
4  
[1, 2, 3, 99]
```

Why: append mutates the list in place to [1,2,3,99]; len is 4; the caller's list is changed.

Q7. Answer: a

```
Output:  
False  
True
```

Why: any of all-zeros is False; all of non-zero numbers is True.

Q8. Answer: b

```
Output:  
True  
False
```

Why: 5 is truthy so any is True; 0 and "" are falsy so all is False.

Q9. Answer: a

```
Output:  
2  
bool
```

Why: True counts as 1, so the sum is 2; an all-boolean array has dtype bool.

Q10. Answer: b

```
Output:  
1  
2
```

Why: argmax/argmin return the INDEX, not the value: first max 7 is at index 1, min 1 is at index 2.

Q11. Answer: b

```
Output:  
[(1, 'x'), (2, 'y')]
```

Why: zip stops at the shortest iterable, so only two pairs — no error, no None padding.

Q12. Answer: a

```
Output:  
ab-ab-ab
```

Why: ["ab"]*3 makes three "ab", joined by "-" gives ab-ab-ab.

Q13. Answer: a

```
Output:  
=====  
(blank line)
```

Why: "="*5 repeats the string to "====="; ""*0 is the empty string "", which prints a blank line. No error.

Q14. Answer: c

```
Output:  
ValueError
```

Why: (3,) + (2,): two 1-D arrays of different lengths can't broadcast, so ValueError.

Q15. Answer: b

```
Output:  
None  
[1, 2, 3]
```

Why: sort() sorts in place and returns None, so x is None and nums becomes [1,2,3].